S&P GLOBAL & CRISIL | CODE TO CONNECT HACKATHON 2026

FinRisk

Explainable Event-Driven Market Risk Intelligence

“From unstructured market events to explainable index rebalancing and stress-testing decisions.”



Explainable | Auditable | Bounded

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From event noise to portfolio decisions

Analysts face hundreds of headlines across news, filings and social feeds. They need fast, auditable signals, not just sentiment scores.

1

Event sources

News (GDELT), SEC EDGAR filings, StockTwits and tweet replay: unstructured text, around the clock.

2

RiskSignal

Sentiment, event type, impact, confidence, provenance and explanation in one frozen schema.

3

Portfolio decision

Module A tilts index weights within bounds. Module B stress-tests the book with Monte Carlo.

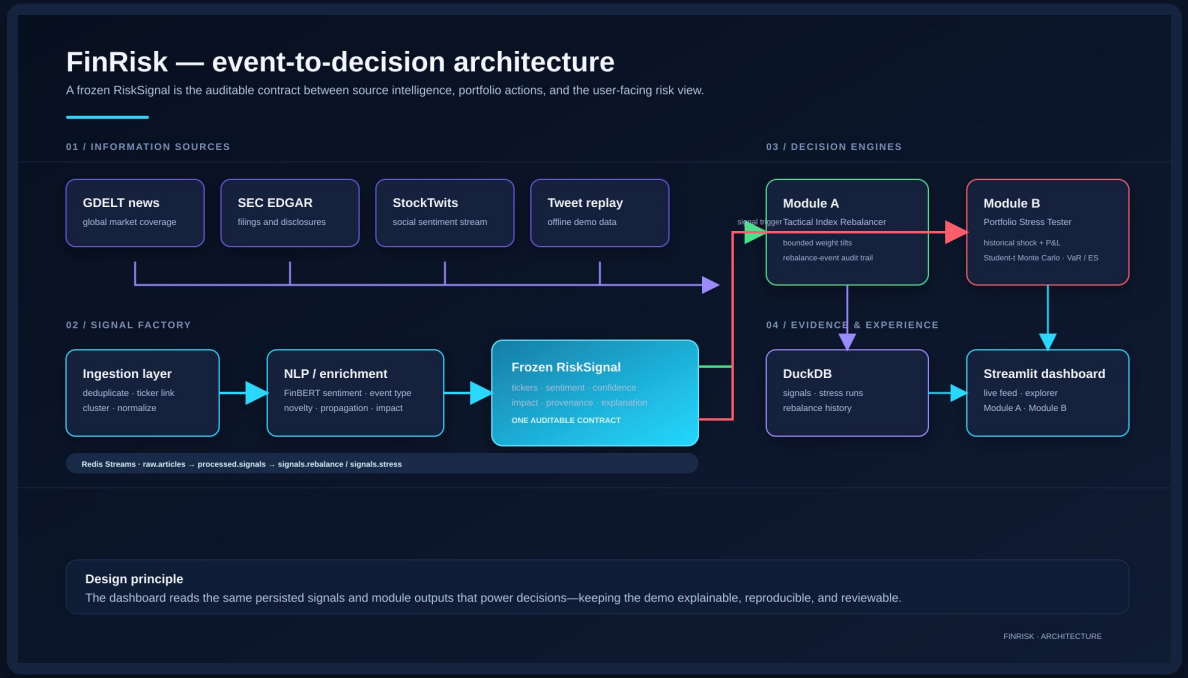
4
data sources

10
event types classified

2
downstream risk modules

330+
automated tests

One signal contract, two risk engines



INGEST

SHA-256 dedup, ticker linking and event clustering across four sources.

UNDERSTAND

FinBERT sentiment, event classifier, XGBoost impact model with rule-based fallback.

CONTRACT

A frozen RiskSignal schema is the single interface for every consumer.

DECIDE AND SERVE

Rebalancer and stress tester; Redis Streams, DuckDB, FastAPI, Streamlit.

Diffusion / propagation score: estimates whether an event is spreading across sources and becoming more material (a Hawkes-lite decay feature, not a generative model).

Every signal explained and auditable



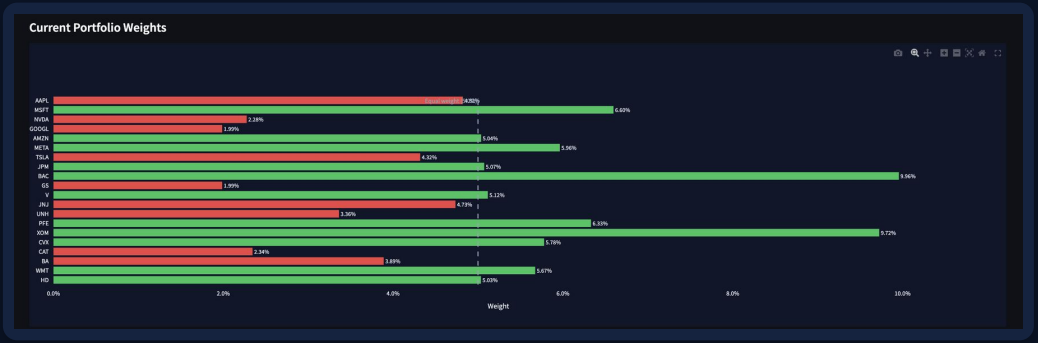


+0.318

Spearman ρ of the impact model, trained on abnormal returns (stock minus SPY). Rule-based baseline: −0.067.

- FinBERT gives financial-domain sentiment
- 10 event types: earnings, M&A, regulatory, geopolitical, credit and more
- Novelty and propagation prioritise emerging, multi-source events
- Immutable RiskSignal keeps source, cluster ID and explanation

Module A: controlled index rebalancing



Recent Rebalance Events

Time (UTC)	Trigger	Top Weight Changes
2026-10-18 13:50:36	Signal: META Earnings impact=6.0	META: +2.17%, XOM: -1.23%, WMT: -1.16%
2026-10-18 13:48:06	Signal: BAC Earnings impact=7.0	BAC: +3.44%, XOM: -0.67%, JPM: -0.56%
2026-10-18 13:47:56	Signal: HD Leadership impact=2.7	HD: -4.57%, AMZN: +1.12%, WMT: +0.80%
2026-10-18 13:46:50	Signal: HD Leadership impact=5.0	HD: +2.57%, TSLA: -0.90%, PFE: -0.66%
2026-10-18 13:46:05	Signal: META M&A impact=2.2	XOM: -2.01%, META: +1.58%, BA: -1.48%
2026-10-18 13:44:50	Signal: HD Commodity impact=3.3	HD: -3.30%, XOM: +1.02%, META: -0.52%
2026-10-18 13:43:50	Signal: TSLA Cyber impact=4.5	TSLA: -2.25%, PFE: -1.46%, XOM: +0.84%
2026-10-18 13:41:04	Signal: AMZN Cyber impact=7.9	XOM: +2.07%, WMT: -1.34%, PFE: -1.32%
2026-10-18 13:38:39	Signal: V Macro impact=6.5	TSLA: +2.09%, V: -1.33%, XOM: +0.72%
2026-10-18 13:38:29	Signal: XOM Credit impact=8.7	XOM: -2.27%, CAT: -0.57%, AAPL: +0.32%

Signal to weight tilt

High-materiality signals become bounded weight adjustments, never a full reallocation.

Reviewable log

Every rebalance records its trigger signal and its top weight changes.

Controlled by design

An equal-weight base stays as the reference. A demo of controlled reaction, not autonomous trading.

Synthetic 20-stock index. Not investment advice.

Module B: stress testing with tail risk

−$1.55M
Deterministic ΔNAV (−2.60%), Credit event, impact 8.0

$3.00M
Monte Carlo VaR 95%

$3.60M
Expected Shortfall 95%

10,000
Student-t (df=5) paths, Cholesky-correlated





- Bonds use duration-convexity, loans use PD x LGD x EAD, swaps use DV01.
- A reverse stress test finds the minimum impact that breaches a loss limit.

Histogram is reconstructed from VaR/ES summary statistics, not raw paths.

Impact, limits and next steps

Impact

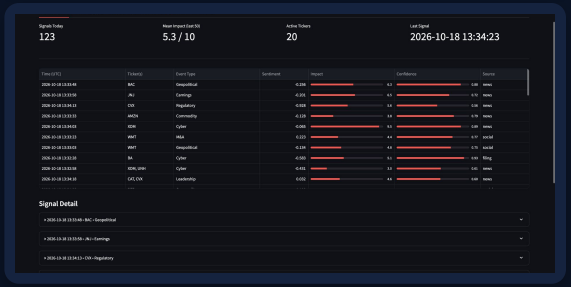
- Event noise becomes explainable, reviewable action
- Every portfolio response traces back to its originating signal
- Runs offline in replay mode with a one-command Docker demo

Limitations

- Synthetic portfolio and signals; not investment advice
- Impact model trained on synthetic prices
- Loss histogram reconstructed from VaR/ES

Next steps

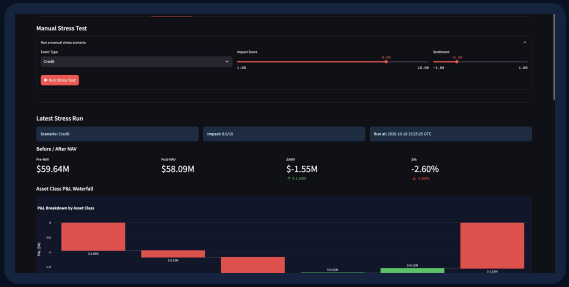
- Real-data evaluation and better calibration
- Stronger monitoring and drift checks
- Broader scenario coverage



Live Risk Feed



Module A: Index Rebalancer



Module B: Stress Tester

GitHub: github.com/MMWARRIOR03/VIT_Vellore-Mrinal_Mohit-hackathon